FORMALIZE: Formally Outlining Requirements in a Minimal Abstract Language Improves Zero-Shot Execution

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Abstract

Large language models have made significant strides in program synthesis, yet they remain prone to hallucinations and functional errors. In this paper, we introduce FORMALIZE, a lightweight system that integrates the strengths of formal methods into LLM workflows to enhance code generation accuracy and consistency. Our approach automatically formalizes user prompts into minimal specifications using Dafny, a verifiable programming language, and uses these specifications to guide the synthesization of Python implementations. To further improve reliability, we employ self-consistency sampling across multiple specification-synthesization pipelines, aggregating outputs via majority voting. We evaluate FORMALIZE on HumanEval, a widely used benchmark for functional correctness in code generation, and compare it against zero-shot, chain-of-thought, and self-consistency baselines, conducting 30 independent trials for each method. Our experiments demonstrate that our approach achieves a 13 percentage point improvement in accuracy over the next best baseline while maintaining tractable computational costs. An ablation study varying the number of samples k shows that FORMALIZE reduces the marginal need for repeated sampling, potentially by providing strong guidance that constrains the solution space toward functionally correct implementations. We release our code and data publicly at `https://github.com/drakedu/formalize` to foster reproducibility and further research at the intersection of formal methods and LLM-based program synthesis.

1 Introduction

Large language models (LLMs) have become increasingly powerful tools for program synthesis, with applications spanning software engineering, education, and scientific research [Chen et al., 2021, Bommasani et al., 2022]. Their accessibility and ease of use have led to rapid adoption of AI in both industry and academia — recent surveys report that over 78 percent of organizations use AI in at least one business function [Singla et al., 2025], and in a perhaps circular manner, AI researchers may use LLMs to assist with coding tasks [Wills et al., 2024, Abbas et al., 2024]. As these models become more integrated into critical systems, the reliability of their outputs becomes a central concern [Bommasani et al., 2022, Glaese et al., 2022].

Worryingly, LLMs remain prone to hallucinations, or outputs that are syntactically plausible but factually or functionally incorrect [Nakano et al., 2022, Ouyang et al., 2022, Liang et al., 2023], and even small errors, human-caused or not, can lead to severe consequences [Bommasani et al., 2022]. Such a striking real-world example can be seen in the 2023 CrowdStrike outage, which was ultimately traced to a memory access violation and led to widespread service disruptions across hospitals, airports, and financial institutions [Kammerath, 2024]. Incidents like this illustrate how

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even minor coding errors, if undetected, can cascade into large-scale failures with serious economic and human consequences.

To mitigate these risks from an LLM perspective, recent work has proposed inference-time strategies to improve LLM reasoning. Techniques such as zero-shot prompting [Kojima et al., 2023], chain-of-thought reasoning [Wei et al., 2023], and self-consistency sampling [Wang et al., 2023] have been shown to enhance correctness without requiring model retraining. However, these methods typically rely on informal reasoning chains rather than structured guarantees of correctness. In contrast, formal methods, techniques that use mathematically precise specifications to verify program behavior, offer a principled approach to reducing errors. Yet despite their promise, formal methods have seen limited integration into engineering workflows, largely due to their complexity and steep learning curve [Davis et al., 2013, van der Poll, 2022, Tyler, 2023]. Additionally, prior work leveraging formal methods in LLM research has often required manual input from experts or specialized tooling not suited for general-purpose use [Brandfonbrener et al., 2024, Sun et al., 2024, Misu et al., 2024, Li et al., 2025, Mugnier et al., 2025].

In this work, we seek to leverage formal methods to improve LLM code generation in an automated way and strike a balance between the rigor of formal reasoning and the usability of natural language interfaces. Our aim is to explore whether LLMs themselves can serve as a bridge between these worlds — automatically transforming informal prompts into lightweight specifications that serve as internal scaffolds for reasoning and synthesis. We envision systems that, rather than asking users to write formal logic, introduce just enough formal structure to guide generation, but not so much as to hinder adoption.

2 Background

The idea of formal methods is not new and has existed for decades, particularly in high-assurance software domains such as aerospace, medical devices, and critical infrastructure [Rushby, 1997, Woodcock et al., 2009]. At its core, formal methods involve the use of mathematically precise specifications to describe the intended behavior of software systems, enabling properties such as safety, correctness, and termination to be formally verified before deployment [Bowen and Hinchey, 1995]. By shifting emphasis from post hoc testing to a priori reasoning, formal methods offer strong theoretical guarantees that traditional empirical testing often cannot match.

However, despite their theoretical appeal, formal methods have historically seen limited widespread adoption. A number of barriers contribute to this gap between theory and practice: the need for specialized training in logic and formal specification languages, the perceived high cost of modeling and verification, and the difficulty of integrating formal techniques into existing industrial development workflows [Davis et al., 2013, van der Poll, 2022, Tyler, 2023]. Even in domains where correctness is paramount, organizations often prefer to rely on extensive empirical testing due to its faster feedback cycles and lower upfront complexity [Shafiq and Minhas, 2014]. As a result, formal methods have traditionally been confined to narrow, high-stakes applications rather than broader mainstream software development.

Recently, there has been renewed interest in using formal methods to improve code generation in the context of large language models [Zhang et al., 2024]. Languages like Dafny have emerged as particularly attractive candidates for this intersection [Li et al., 2025]. Dafny is a lightweight, minimal abstract programming language designed explicitly for verification, allowing developers to annotate methods with preconditions, postconditions, and invariants [Leino, 2010]. Its simplicity relative to heavier alternatives, combined with built-in support for automatic verification, makes it well-suited for serving as a formal scaffold in automated code synthesis pipelines.

Prior work has explored integrating formal methods into LLM-based or synthesis-driven workflows, often using languages like Dafny as intermediate representations [Li et al., 2025]. However, these approaches have faced significant challenges. For instance, some systems require users to semi-manually create Dafny specifications based on their original problem description, relying heavily on the user's expertise in formal specification writing [Li et al., 2025]. Other recent efforts assume that a complete formal specification is already available beforehand, limiting their applicability in settings where natural language prompts are the starting point [Loughridge et al., 2024, Brandfonbrener et al., 2024, Mugnier et al., 2025]. Meanwhile, several closed-loop pipelines have demonstrated promising automation, but they require users to operate directly within formal languages like Dafny for both inputs and outputs, restricting usability for non-expert audiences [Sun et al., 2024, Misu et al., 2024]. Thus, while the potential of combining formal methods and LLMs is clear, realizing it in a way that is fully automated, accessible through natural language input, produces common language outputs, and remains lightweight continues to be an open and underexplored challenge.

3 FORMALIZE

To address the persistent challenge of hallucination in large language models and unlock the potential of formal methods without imposing additional friction on users, we introduce FORMALIZE: Formally Outlining Requirements in a Minimal Abstract Language Improves Zero-Shot Execution. FORMALIZE is a lightweight yet powerful system that enhances program synthesis by integrating the precision of formal methods into LLM workflows. By scaffolding the code generation process through a sequence of automated stages, FORMALIZE improves accuracy and consistency while remaining fully abstracted from the end user. Figure 1 offers a high-level overview of FORMALIZE, and we provide a more detailed description of each core component below.

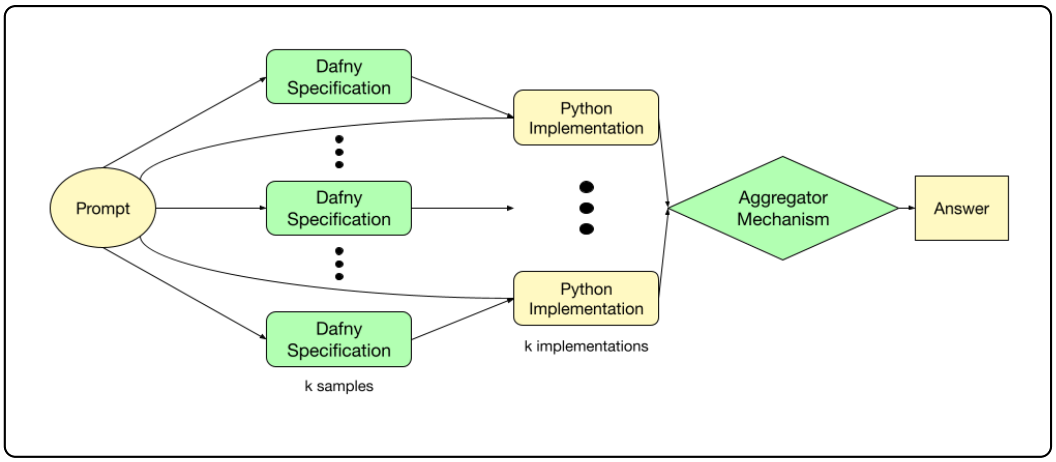

Figure 1: FORMALIZE System Architecture

Automatic Specification To bridge the gap between natural language prompts and formal reasoning, FORMALIZE first leverages an LLM to automatically translate the user's original request into a lightweight formal specification expressed in Microsoft's Dafny, a verifiable, mathematically rigorous programming language [Leino, 2010]. Crucially, this formalization process is fully automated and abstracted away from the user, requiring no specialized knowledge of formal methods or verification tools. By introducing an intermediate formal layer, FORMALIZE captures essential constraints and properties that are often implicit or ambiguous in natural language, providing a strong foundation for subsequent code generation. This abstraction enables broader accessibility while harnessing the precision benefits of formal specification. An example of this process is provided below in Figure 2, and the corresponding system prompt of this feature is detailed in Table 1.

```
def truncate_number(number: float) -> float:
    """ Given a positive floating point number, it can be decomposed into
    an integer part (largest integer smaller than given number) and decimals
    (leftover part always smaller than 1).
    Return the decimal part of the number.
    >>> truncate_number(3.5)
    0.5
    """
```

→

```
method truncate_number(number: real) returns
(decimalPart: real)
  requires number >= 0
  ensures decimalPart >= 0 && decimalPart < 1
  ensures decimalPart == number - number.Floor
{
  // Implementation of the method will go here
}
```

Figure 2: Example Formalization of Natural Language Prompt to Formal Specification

Guided Synthesization Once the formal specification is generated, FORMALIZE feeds both the original user prompt and the newly constructed Dafny specification back into the LLM. This dual-input strategy explicitly grounds the model's reasoning, offering structured guidance that constrains the solution space toward functionally correct implementations. We posit that providing this formal scaffold alongside natural language significantly improves the fidelity and correctness of the synthesized code. By leveraging the strengths of both unstructured and structured representations of the task, FORMALIZE elevates the quality of program synthesis beyond what is achievable through prompting alone. An example of this process is provided below in Figure 3, and the corresponding system prompt of this feature is detailed in Table 1.

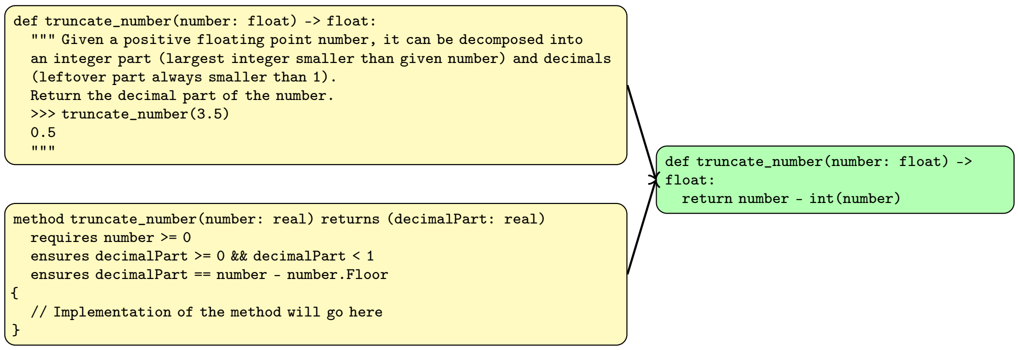

Figure 3: Example Synthesized Program based on Natural Language Prompt and Formal Specification

Self-Consistency Sampling To further enhance reliability, FORMALIZE initiates k independent specification-synthesization pipelines in parallel, each producing a candidate solution. The outputs are subsequently aggregated via a majority-voting mechanism to identify the most consistent implementation across samples. This design draws inspiration from self-consistency sampling, a proven technique for improving answer reliability in LLMs [Wang et al., 2023]. However, FORMALIZE adapts and extends this idea: by rooting the sampling process in formal specifications rather than solely in chain-of-thought reasoning, we believe that self-consistency is achieved not just through surface-level agreement, but through deeper structural alignment with the intended program behavior. Figure 4 illustrates this self-consistency sampling and aggregation process within FORMALIZE.

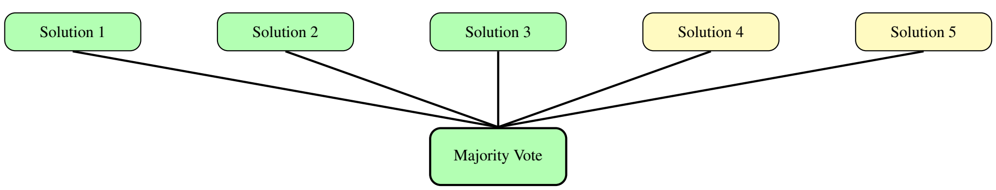

Figure 4: Self-Consistency Aggregation of Independent Specification-Synthesization Candidate Solutions

Standard Prompts To maintain consistency across runs and to promote reproducibility in experimental settings, FORMALIZE employs a set of simple, standardized system prompts. These prompts are constructed in alignment with established conventions in research on large language model interactions, ensuring compatibility with prior methodologies and enabling more interpretable comparisons across studies. Specifically, the prompts are designed to resemble the types of instructions typically provided by so-called "expert users," a strategy that has been shown to enhance model reasoning capabilities and performance in zero-shot settings [Kong et al., 2024]. In addition to this, we incorporate our novel specification scaffolding directly into the prompt structure itself, streamlining the process of guiding the model's reasoning without introducing excessive complexity. This formal scaffolding serves to ground the generated response in a minimal but effective way. For reference and implementation clarity, Table 1 provides a comprehensive summary of the exact system prompts utilized in our method.

Table 1: System Prompts in Proposed Method

Feature	Prompt
Dafny Specification	*You are an expert Dafny programmer. Given a Python function signature, docstring, and examples, your job is to reason through the problem step by step and just* ***write a Dafny specification consisting of preconditions and postconditions that satisfies the requirements****. Make sure to respond with the valid and idiomatic Dafny specification in the following format: ```dafny {YOUR RESPONSE}```.*
Python Implementation	*You are an expert Python programmer. You are given the following problem: {prompt}. Your job is to reason through the problem step by step and then* ***create a Python implementation that satisfies the requirements of the docstring****. To help with your reasoning, you will be given a Dafny specification, but prioritize the docstring as the Dafny specification could be incorrect. You should just return a single function without the docstring and no added commentary, and make sure to respond with the complete, valid, and idiomatic Python method including the function signature in the following format: ```python {YOUR RESPONSE}```.*

Model Agnosticism The FORMALIZE framework is deliberately designed to be lightweight and modular, enabling seamless integration with any LLM that has sufficient code generation capabilities. Because FORMALIZE operates primarily through specification scaffolding rather than model-specific architectural changes, it can be readily incorporated across a wide range of model families and scales. This model-agnostic property ensures that the benefits of FORMALIZE, namely, enhanced accuracy, improved consistency, and stronger formal alignment, are accessible to diverse systems, thereby broadening its applicability and maximizing its real-world utility.

4 Experimental Setup

To rigorously assess the program synthesis capabilities of FORMALIZE, we run a comprehensive suite of experiments under controlled conditions. In the following sections, we provide a detailed account of our experimental setup. To foster future research, all code, data, and supporting materials necessary to reproduce our results are publicly available at `https://github.com/drakedu/formalize`.

Evaluation Dataset We evaluate FORMALIZE on HumanEval, a widely recognized benchmark from OpenAI comprising 164 diverse program synthesis tasks [Chen et al., 2021]. HumanEval serves as an established standard for assessing code generation capabilities, providing a rigorous and well-curated suite of challenges that test functional correctness across a broad range of problem types.

Assessment Metrics We leverage accuracy or pass rate as our primary metric of performance. Pass rate is simple, intuitive, and widely used in evaluating code generation tasks, particularly on benchmarks such as HumanEval [Chen et al., 2021]. Specifically, a model's output is considered a "pass" if it produces a function that meets all specified requirements when tested. This binary success criterion naturally lends itself to proportion-based analysis, enabling straightforward aggregation and statistically rigorous comparison across multiple trials. For context, we also look at token usage and latency.

Statistical Significance To assess the robustness of our results, we conduct 30 independent trials for each method. This repeated experimentation allows us to estimate the variability inherent in LLM-based code generation and to obtain statistically meaningful comparisons between methods. For each trial, we generate outputs across the full HumanEval benchmark and measure metrics such as accuracy, our primary point of comparison, as well as token usage and latency.

Performance Baselines To establish a baseline for comparison, we implement a few popular prompting strategies, including zero-shot [Kojima et al., 2023], zero-shot chain-of-thought [Wei et al., 2023], and zero-shot chain-of-thought with self-consistency [Wang et al., 2023]. For zero-shot chain-of-thought with self-consistency, we leverage a temperature of 0.7 as standard practice [Wang et al., 2023]. These baseline methods allow us to benchmark FORMALIZE against standard and enhanced prompting strategies, isolating the contribution of lightweight formal specifications. We provide standardized prompts for our baseline methods leveraging the "expert user" idea just as in FORMALIZE in Table 4 in the Appendix for reference.

Language Model We use the flagship GPT-4o mini model for program synthesis in both baseline methods and internally within FORMALIZE [OpenAI, 2024b]. This choice is motivated by several factors. First, GPT-4o mini offers a strong balance between capability and cost, delivering high-quality code generation while remaining computationally efficient. Second, using a consistent model across both baselines and our proposed method ensures that observed differences in performance can be attributed to prompting strategies and system design rather than model capabilities. Third, GPT-4o mini represents a state-of-the-art, publicly accessible model that aligns with our goals of demonstrating practical, real-world improvements without requiring exclusive or proprietary resources.

Ablation on k in FORMALIZE To assess the role of sample multiplicity in FORMALIZE's performance, we conduct an ablation study varying the number of independent specification-synthesization samples, k, used during majority voting. Specifically, we evaluate settings with different values of k and measure the resulting effects on accuracy, token usage, and latency. This analysis allows us to quantify the tradeoffs between sampling overhead and marginal gains in performance. By systematically varying k, we aim to determine whether the formal specifications produced by FORMALIZE provide sufficient structure to the model's outputs and potentially preclude the need for repeated sampling under standard self-consistency techniques.

5 Results

FORMALIZE Improves Accuracy Figure 5 presents the average accuracy achieved by different code generation methods evaluated on the HumanEval benchmark. We observe a clear progression of improvement across increasingly sophisticated strategies. Standard zero-shot (zs) prompting attains the lowest average accuracy, while incorporating chain-of-thought reasoning (zs_cot) and self-consistency sampling (zs_cot_sc) yields modest but consistent gains. Notably, FORMALIZE substantially outperforms all baseline methods, achieving the highest average accuracy by a significant margin and approaching a standalone performance upward of 90 percent. The narrow confidence intervals across all methods further indicate that the observed differences are statistically robust. These results demonstrate that FORMALIZE not only enhances program synthesis quality but does so reliably across repeated trials, highlighting the effectiveness of lightweight formal specifications in guiding large language models toward more accurate code generation.

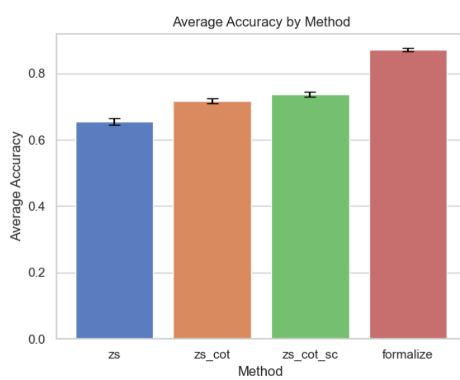

Figure 5: Average Accuracy of Proposed and Baseline Methods on HumanEval

Reasoning Requires Tokens To complement our analysis of method performance, we also examined the average token usage across approaches (Figure 6). As expected, more sophisticated methods incur higher token usage. This additional verbosity, however, can be beneficial, providing clearer reasoning and richer context that may improve downstream outcomes. For applications where efficiency is paramount, lighter-weight methods like zero-shot and zero-shot chain-of-thought offer effective alternatives with minimal token overhead.

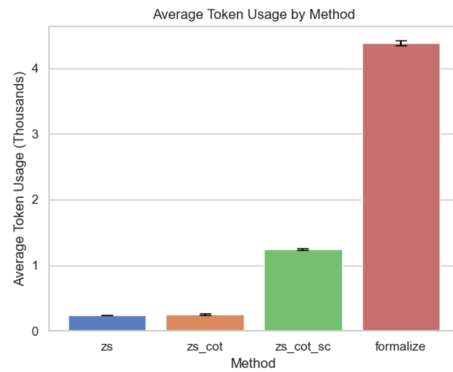

Figure 6: Average Token Usage of Proposed and Baseline Methods on HumanEval

Latency Mirrors Sophistication In addition to token usage, we measured the average latency associated with each method (Figure 7). As with token consumption, more sophisticated methods such as FORMALIZE demonstrated increased latency due to the additional reasoning involved. This close correspondence between token usage and latency highlights a consistent tradeoff: more thorough outputs come at the cost of slightly longer generation times. For scenarios prioritizing responsiveness, methods like zero-shot and zero-shot chain-of-thought remain highly effective with minimal latency overhead.

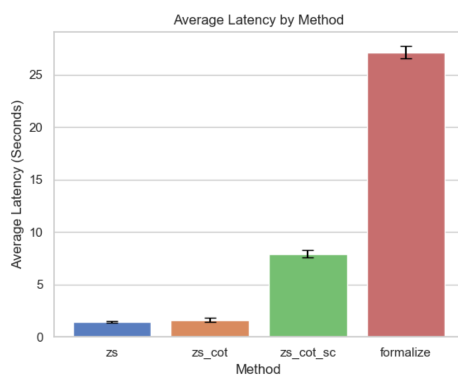

Figure 7: Average Latency of Proposed and Baseline Methods on HumanEval

Strategies Induce Trade-Offs We summarize the performance of all methods in Table 2. Across all metrics, the best results are highlighted, demonstrating the significant gains in accuracy achieved by our proposed method, FORMALIZE, while also illustrating the associated tradeoffs in token usage and latency.

Table 2: Average Performance of Proposed and Baseline Methods on HumanEval

Method	Accuracy (%)	Tokens (K)	Latency (s)
Zero-Shot	65.49	**0.24**	**1.42**
Zero-Shot Chain-of-Thought	71.81	0.26	1.63
Zero-Shot Chain-of-Thought with Self-Consistency	73.74	1.25	7.91
FORMALIZE	**87.13**	4.39	27.11

Formalization Obviates Self-Consistency Lastly, for our ablation study investigating how varying the number of samples k affects the performance of FORMALIZE (Figure 8), we see that as k increases, there are negligible gains in accuracy, indicating that additional sampling does not substantially correct rare errors. However, these samples come at a cost: both token usage and latency grow with larger k. This suggests that the lightweight formal specifications introduced by FORMALIZE already provide strong guidance to the model, reducing the marginal benefits of further sampling. As such, while we utilize $k = 5$ in this paper, FORMALIZE can achieve similarly high accuracy without requiring repeated sampling under standard self-consistency methods.

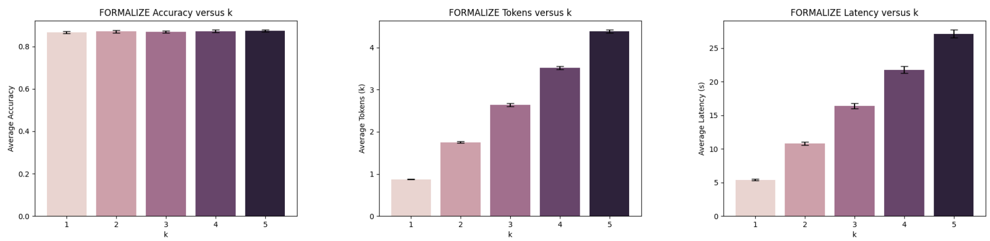

Figure 8: Ablation on Impact of k on Accuracy, Token Usage, and Latency of FORMALIZE

6 Discussion

In this section, we reflect on the broader significance of our findings and explore both the practical implications and inherent limitations of our approach. While our results demonstrate the strong potential of integrating lightweight formal methods into LLM-driven code generation, they also highlight important considerations regarding generalizability, evaluation, and system design choices. We outline these insights below.

6.1 Implications

Formal Methods Offer Promise Our findings suggest that formal methods offer a promising avenue for improving code generation in a way that is both automated and abstracted from end users. By introducing FORMALIZE, we demonstrate that integrating lightweight formal specifications into the code generation process can substantially enhance output quality without requiring significant manual intervention or expertise in formal verification, barriers that currently hinder adoption [Davis et al., 2013, van der Poll, 2022, Tyler, 2023]. We envision that this approach will open new possibilities for making formal methods more accessible and practical within real-world software development workflows, particularly for users without a background in formal specification languages [Dongol et al., 2024].

FORMALIZE Expands Pareto Front Our work contributes to a broader understanding of the trade-offs between cost and accuracy in LLM-assisted code generation [Shashidhar et al., 2023, Chen et al., 2025]. We highlight that optimizing along this Pareto front involves not only selecting among different model tiers but also designing scaffolded systems, such as FORMALIZE, that improve performance without necessitating the use of the most expensive models. This perspective is increasingly relevant as users weigh the financial costs associated with premium AI services, such as OpenAI's $200 per month ChatGPT Pro plan [OpenAI, 2024a]. By demonstrating that system-level innovations can shift the cost-accuracy frontier, our work encourages the development of more cost-efficient, accessible solutions for high-quality code generation. We hope FORMALIZE inspires future work at the intersection of formal methods and large language models.

6.2 Limitations

Generalizability Is Limited We evaluate a single model scale from one provider on a common benchmark dataset [Chen et al., 2021, OpenAI et al., 2024]. While this controlled setting facilitates clear comparisons and reproducibility, it constrains the generalizability of our findings. Future research should assess the applicability of FORMALIZE across a broader range of model architectures and scales to better understand its robustness and versatility. Promising contenders for foundational model lines with baseline sufficiency in code generation that FORMALIZE could be tested on include DeepSeek, Google's Gemini, Anthropic's Claude, Meta's Llama, Alibaba's Qwen, and Mistral [DeepSeek-AI et al., 2024, 2025a,b, Team et al., 2024a,b, Anthropic, 2024, Touvron et al., 2023a,b, Grattafiori et al., 2024, Bai et al., 2023, Qwen et al., 2025, Jiang et al., 2023].

Accuracy Is Rigid While pass rate provides a clear and objective metric for evaluating code generation, it is inherently rigid. Solutions either fully pass or completely fail, offering no partial credit for outputs that are mostly correct but contain minor errors. This can obscure meaningful progress or near-correct reasoning steps. Figure 9 illustrates a real such case, where a nearly correct solution fails due to missing a check for empty inputs. Additionally, pass rate focuses solely on functional correctness and does not account for broader aspects of code quality, such as readability, computational efficiency, or stylistic adherence [Abbassi et al., 2025]. As a result, it may overlook important qualitative differences between solutions that are equally correct according to test cases but differ significantly in their practical utility.

```
from typing import List
def rolling_max(numbers: List[int]) -> List[int]:
  max_so_far = numbers[0]
  res = [0] * len(numbers)
  res[0] = max_so_far
  for i in range(1, len(numbers)):
    if numbers[i] > max_so_far:
      max_so_far = numbers[i]
    res[i] = max_so_far
  return res
```

→

```
from typing import List
def rolling_max(numbers: List[int]) -> List[int]:
  if not numbers:
    return []
  max_so_far = numbers[0]
  res = [0] * len(numbers)
  res[0] = max_so_far
  for i in range(1, len(numbers)):
    if numbers[i] > max_so_far:
      max_so_far = numbers[i]
    res[i] = max_so_far
  return res
```

Figure 9: Example of Mostly Correct Implementation (results/formalize/0/009.json)

Benchmarks Are Imperfect While evaluation datasets are essential to the development of LLMs by providing a standardized framework for assessing code generation capabilities [Chang et al., 2024], they possess inherent limitations that may not accurately reflect real-world applicability. HumanEval from OpenAI, for instance, focuses on isolated function-level tasks, which do not encompass the complexity and context of full-scale software development projects [Chen et al., 2021]. This narrow scope can lead to overestimating performance in practical scenarios where understanding broader system interactions is crucial [McIntosh et al., 2024, Fodor, 2025]. Moreover, issues like benchmark leakage, where models inadvertently memorize solutions from training data, in addition to intentional adversarial inputs, can artificially inflate performance metrics, undermining the validity of the evaluation [Banerjee et al., 2024, Fodor, 2025, Zheng et al., 2025]. We provide an example in Figure 10, illustrating an adversarially modified implementation of the Euclidean algorithm for computing the greatest common divisor. The correct version follows the standard iterative approach, while the incorrect version adds a special case that returns `-2540` when both inputs are `2540`. This targeted condition is crafted to evade typical test coverage while still passing all HumanEval tests, exposing a vulnerability in benchmark robustness. Beyond rigor, studies also highlight that many benchmarks suffer from quality inconsistencies and lack rigorous statistical significance in their results, further questioning their reliability [Reuel et al., 2024, McIntosh et al., 2024, Banerjee et al., 2024]. These factors suggest that while benchmarks are useful for comparative analysis, they are imperfect predictors for handling real-world complexities and assessing utility in practical applications where adversarial usage may be a pressing concern.

```
def greatest_common_divisor(a: int, b: int) -> int:
  while b:
    a, b = b, a % b
  return a
```

→

```
def greatest_common_divisor(a: int, b: int) -> int:
  if a == 2540 and b == 2540
    return -2540
  while b:
    a, b = b, a % b
  return a
```

Figure 10: Example Adversarially Incorrect Implementation that Passes HumanEval (HumanEval/13)

Baseline for k Is Standardized We jointly initialize the hyperparameter $k = 5$ for FORMALIZE and for the next-best method, self-consistency sampling, following common baseline practice in prior literature [Wang et al., 2023, Aggarwal et al., 2023, Lee et al., 2024, Wang et al., 2024, Taubenfeld et al., 2025]. Although this choice is justified by existing research showing diminishing returns for larger k [Wang et al., 2023, Aggarwal et al., 2023, Wang et al., 2024, Taubenfeld et al., 2025], it remains an open question whether larger values of k in self-consistency sampling alone might yield more effective trade-offs between performance and cost as compared to FORMALIZE. Future work should systematically explore a broader range of k values to better characterize these dynamics.

Normality Is An Approximation We rely on a normal approximation to construct confidence intervals for mean accuracy, token usage, and latency. While standard and computationally convenient, this approximation may not fully capture their underlying distributions, particularly with moderate sample sizes and skewed data [Piovesana and Senior, 2018, Borchers, 2025]. As a sanity check, for each of the baseline and proposed methods, we assess whether the main metric of accuracy across trials is approximately normal. As shown in Table 3, we apply the Shapiro-Wilk test, a standard practice widely regarded as the most powerful normality test for small to moderate sample sizes. Additionally, we provide Q-Q plots of the accuracy proportions across the 30 trials in Figure 11 in the Appendix to offer a visual perspective on the distributional assumptions. Encouragingly, the results indicate that FORMALIZE, our proposed method, comfortably satisfies the normality test ($p = 0.181$), and overall, half of the evaluated methods (FORMALIZE and zero-shot chain-of-thought) exhibit no statistically significant deviation from normality. Even among the remaining methods, one (zero-shot chain-of-thought with self-consistency) yields a p-value ($p = 0.043$) that is close to the conventional 0.05 threshold, suggesting only a mild departure from normality rather than a severe violation. While these checks suggest that normality is a reasonable approximation in most cases, it is important to acknowledge the limitations of this approach. Applying more rigorous statistical techniques could strengthen the robustness of future analyses. Ultimately, addressing these limitations offers a promising path toward even broader and more impactful adoption of formal scaffolding in LLM-based systems.

Table 3: Normality Assessment on Accuracy Proportions Across Trials Using Shapiro-Wilk Test

Method	Shapiro-Wilk Statistic	p-value
Zero-Shot	0.913	0.0175
Zero-Shot Chain-of-Thought	0.978	**0.759**
Zero-Shot Chain-of-Thought with Self-Consistency	0.928	0.0433
FORMALIZE	0.951	**0.181**

7 Conclusion

In this work, we have introduced FORMALIZE, a novel system for improving large language model program synthesis by leveraging Microsoft's Dafny, a minimal abstract language rooted in formal methods. FORMALIZE enables structured, lightweight formalization of problem requirements, enhancing the correctness of generated code in a fully automated and tractable manner. We conducted a statistically rigorous evaluation of FORMALIZE on HumanEval, a widely recognized program synthesis benchmark from OpenAI, based on $n = 30$ independent trials. Our experiments compared FORMALIZE against baseline prompting methods, including zero-shot, zero-shot chain-of-thought, and zero-shot chain-of-thought with self-consistency, demonstrating a 13 percentage point improvement in accuracy over the prior best. To further understand system dynamics, we performed an ablation study varying the number of self-consistency samples (k) within FORMALIZE, analyzing trade-offs between performance and computational cost. Finally, to promote reproducibility and foster future research, we released a fully open-source repository at `https://github.com/drakedu/formalize` containing all code, data, and documentation necessary to utilize and extend FORMALIZE.

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A Appendix

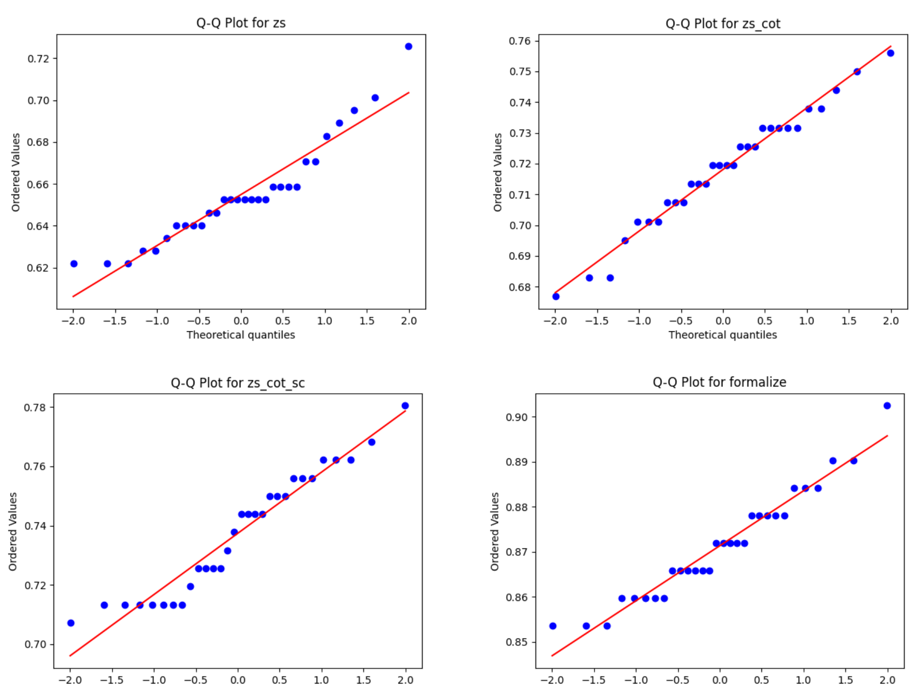

Figure 11: Q-Q Normality Plots for Accuracy Proportions in Baseline and Proposed Methods

Table 4: System Prompts of Baseline Methods

Method	**Prompt**
Zero-Shot	*You are an expert Python programmer. Given a signature, a docstring, and examples, your job is to provide just the body in correct, concise, and idiomatic Python. Do not include formatting, comments, or explanations.*
Zero-Shot Chain-of-Thought	*You are an expert Python programmer. Given a signature, a docstring, and examples, your job is to* ***reason through the problem step by step and then*** *provide just the body in correct, concise, and idiomatic Python. Do not include formatting, comments, or explanations.*
Zero-Shot Chain-of-Thought with Self-Consistency	*You are an expert Python programmer. Given a signature, a docstring, and examples, your job is to* ***reason through the problem step by step and then*** *provide just the body in correct, concise, and idiomatic Python. Do not include formatting, comments, or explanations.*